

C37_A4

Changze Wu

December 2024

1 Q1

1.1 a

Let $g : [a, b] \rightarrow [a, b]$ be a continuous function on the interval $[a, b]$, and assume: g is differentiable on (a, b) and there exist a constant $L, 0 \leq L < 1$, such that $\|g'(x)\| \leq L$ for all $x \in (a, b)$. Then, we can get that $g(x)$ has a unique fixed point $\tilde{x} \in [a, b]$, where $g(\tilde{x}) = \tilde{x}$. And, we can also get that for any initial point $x_0 \in [a, b]$, the sequence defined by $x_{k+1} = g(x_k)$ converges to $\tilde{x}$.

1.2 b

Start with $x_0 \in [a, b]$ and iterate with $x_{k+1} = g(x_k)$, $k = 1, 2, \dots$, then all $x_k \in [a, b]$. Moreover, $x_{k+1} - x_k = g(x_k) - g(x_{k-1}) = g'(x_c)(x_k - x_{k-1})$, for some $x_c \in [x_{k-1}, x_k]$, by MVT.

Thus, $|x_{k+1} - x_k| \leq g'(x_c)(x_k - x_{k-1}) \leq L(x_k - x_{k-1})$

Then, $L|x_k - x_{k-1}| \leq L^2|x_{k-1} - x_{k-2}| \leq \dots \leq L^k|x_1 - x_0|$, Since $L < 1$, $|x_{k+1} - x_k| \rightarrow 0$ as $k \rightarrow \infty$

Hence, x_k converge to some point $\tilde{x} \in [a, b]$

1.3 c

we need to show $g(\tilde{x}) = \tilde{x}$

1 Since x_k converges to $\tilde{x}$, we have $x_k \rightarrow \tilde{x}$ and $x_{k+1} \rightarrow \tilde{x}$

2. By the continuity of g , taking the limit in the iterative relation $x_{k+1} = g(x_k)$ gives: $\tilde{x} = g(\tilde{x})$.

Thus, $\tilde{x}$ is a fixed point of $g(x)$

1.4 d

Prove by contradiction, assume there are two fixed point x_1 and x_2 which are all $\in [a, b]$ such that $g(x_1) = x_1$ and $g(x_2) = x_2$.

By MVT, we get that there exist a $x_c \in [x_1, x_2]$ such that $g(x_2) - g(x_1) = g'(x_c)(x_2 - x_1)$

Since x_1, x_2 are fixed points, $g(x_1) = x_1, g(x_2) = x_2$, by substituting we can get $x_2 - x_1 = g'(x_c)(x_2 - x_1)$

Factoring out that $x_2 - x_1 \neq 0$, then we get $1 = g'(c)$ which contradict that $g'(x_c) \leq L < 1$.

QED.

2 Q2

2.1 a

we analyze the convergence of each iterative method near the root $x = 2$

1. For schema 1, when x approach to 0, the iteration go diverge because it approaches to ∞ . Near $x = 2$, the derivative of $g(x)$ would need to satisfy $|g'(x)| < 1$ to ensure convergence. However, the terms x_k^{-4} and x_k^{-1} make $g'(x)$ grow rapidly for small x_k , causing potential divergence.

Thus, the schema 1 is unable and cannot reliably to solve the equation

2. this schema is stable because the denominator is always positive since $x > 0$. Then we check the convergence near $x = 2$ compute $g'(x) = 130^{\frac{1}{3}} \times \frac{-1}{3} \times (x^2 + 12)^{\frac{-4}{3}} \times 2x$, when $x = 2$, the $|g'(x)| < 1$. Thus could work and is stable near $x = 2$.

3. This scheme is derived from a Newton-like iteration formula, which typically converges quadratically. When $x = 2$, the numerator and denominator behave good and the iteration converges quickly as long as x_k is not too far from $x = 2$. Thus, this one is well-performed and convergent.

2.2 b

The best choice is the third one because it is derived from Newton's method, which has quadratic convergence near the root. This means the error decreases much faster with each iteration compared to second one, which has linear convergence. Although the second one also converges, it is slower and less efficient than third one.

2.3 c

we derive this by Newton's method. We suppose that $f(x) = x^5 + 12x^3 - 130$. Applying the Newton Formula : $x_{k+1} = x_k - \frac{f(x_k)}{f'(x)}$. $f'(x) = 5x^4 + 36x^2$, then we get $x_{k+1} = x_k - \frac{x_k^5 + 12x_k^3 - 130}{5x_k^4 + 36x_k^2}$.

This method is good because it is based on the full derivative of $f(x)$, ensuring faster (quadratic) convergence near the root. In the meanwhile, this is the direct application of Newton's method and works efficiently if the initial guess x_0 is sufficiently close to the root but the third one in question is modified one.

thus, the final one is $x_{k+1} = \frac{4x_k^5 + 36x_k^3 + 130}{5x_k^4 + 36x_k^2}$.

3 Q3

1. we defined a error $e_k = x_k - a^{\frac{1}{3}}$, then we substituted $x_k = e_k + a^{\frac{1}{3}}$.

Then we substituted into the term using Taylor expansions

$$x_k^{-2} = \frac{1}{(a^{1/3} + e_k)^2} = \frac{1}{a^{\frac{1}{3}} + 2a^{\frac{1}{3}}e_k + e_k^2} \approx \frac{1}{a^{2/3}} \left(1 - \frac{2e_k}{a^{1/3}} + \frac{3e_k^2}{a^{2/3}} \right).$$

$$x_k^{-5} = \frac{1}{(a^{1/3} + e_k)^5} \approx \frac{1}{a^{5/3}} \left(1 - \frac{5e_k}{a^{1/3}} + \frac{15e_k^2}{a^{2/3}} \right).$$

Then we substituted them into the final one : then we get $x_{k+1} = p(a^{1/3} + e_k) + \frac{qa}{a^{2/3}} \left(1 - \frac{2e_k}{a^{1/3}} + \frac{3e_k^2}{a^{2/3}} \right) + \frac{ra^2}{a^{5/3}} \left(1 - \frac{5e_k}{a^{1/3}} + \frac{15e_k^2}{a^{2/3}} \right).$

by separating and reordering the term : $x_{k+1} = a^{1/3}(p + q + r) + e_k(p - \frac{2q}{a^{1/3}} - \frac{5r}{a^{1/3}}) + e_k^2(\frac{3q}{a^{2/3}} + \frac{15r}{a^{2/3}}) + \dots$

then we can get three equivalents :

1. $p + q + r = 1$
2. $p - \frac{2q}{a^{1/3}} - \frac{5r}{a^{1/3}} = 0.$
3. $\frac{3q}{a^{2/3}} + \frac{15r}{a^{2/3}} = 0$

then we get that $r = \frac{a^{\frac{1}{3}}}{-4a^{\frac{1}{3}} - 5}, q = \frac{5a^{\frac{1}{3}}}{4a^{\frac{1}{3}} + 5}, p = \frac{5}{4a^{\frac{1}{3}} + 5}$

For the optimized values of p , q , and r , the linear error term is eliminated, leaving: $e_{k+1} \propto e_k^2$. which indicates that quadratic convergence, where the error decreases as the square of the previous error, leading to much faster convergence compared to linear methods.

4 Q4

4.1 a

suppose $p(x) = c_0 + c_1x + c_2x^2$.

according to the question, we have :

$$\begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ 12 \end{bmatrix}.$$

we solve the system : $c_0 - c_1 + c_2 = 4, c_0 = 6, c_0 + c_1 + c_2 = 12 \implies c_1 + c_2 = 6.$
 $c_0 - c_1 + c_2 = 4 \implies c_1 - c_2 = 2$

then we get $c_0 = 6, c_1 = 4, c_2 = 2$ which implies the $p(x) = 6 + 4x + 2x^2$

4.2 b

we apply the lagrange form of the interpolating poly: $p(x) = \sum_{i=0}^n y_i L_i(x)$
where $L_i(x) = \prod_{j \neq i} \frac{x-x_j}{x_i-x_j}$

then we compute for $i = 0, 1, 2$: $L_0(x) = \frac{x-0}{-1-0} \cdot \frac{x-1}{-1-1} = -x \cdot \frac{x-1}{2} = \frac{x^2-x}{2}$

$L_1(x) = \frac{x-(-1)}{0-(-1)} \cdot \frac{x-1}{0-1} = (x+1) \cdot -(x-1) = -(x^2-1)$

$L_2(x) = \frac{x-(-1)}{1-(-1)} \cdot \frac{x-0}{1-0} = \frac{x+1}{2} \cdot x = \frac{x^2+x}{2}$

then we combine them into the final one : $p(x) = 4L_0(x) + 6L_1(x) + 12L_2(x)$

$p(x) = 2(x^2-x) - 6(x^2-1) + 6(x^2+x)$

then we get the final answer same with a after factoring : $p(x) = 6 + 4x + 2x^2$

4.3 c

recalling the Newton form : $p(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1)$

first we calculated $p(-1) = 4 \rightarrow a_0 = 4$ for $x_0 = -1$

then we do next $p(0) = 6 \rightarrow 6 = 4 + a_1(0 - (-1)) \rightarrow a_1 = 2$,

then we do next $p(1) = 12 \rightarrow 12 = 4 + 2(1 - (-1)) + a_2(1 - (-1))(1 - 0) \rightarrow$
 $12 = 4 + 2 \cdot 2 + a_2 \cdot 2 \cdot 1 \rightarrow a_2 = 2$

Thus, the polynomial is: $p(x) = 4 + 2(x + 1) + 2(x + 1)x \Rightarrow p(x) = 6 + 4x + 2x^2$

4.4 d

in each question i changed them into monomial basis and find all of them are equal to $p(x) = 6 + 4x + 2x^2$, thus we can easily get that they are the same.

4.5 e

1. for the first method, we need to get the Vandermonde system first: then we constructing the Vandermonde matrix requires $O(n^2)$ flops because there are $(n+1) \times (n+1)$ entries, and each requires at most n multiplications for the powers. then we solve them using Gaussian elimination will **cost** $O(n^3)$

2. for the second methods, we computing each $L_i(x)$ involves a product over n terms (excluding $j = i$), and each product requires $O(n)$ multiplications. Since there are $n+1$ basis polynomials, constructing all $L_i(x)$ requires $O(n^2)$ by the way, Multiplying each $L_i(x)$ by y_i and summing to form $p(x)$ requires an additional $O(n^2)$ flops. **thus** we cost total $O(n^2)$

3. the newtome expression is like $p(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + \dots + a_n(x - x_0)(x - x_1) \dots (x - x_{n-1})$, where the coefficient are determined by solving a lower triangular system based on the interpolation conditions.

Each equation involves at most $i+1$ terms (for $i = 0, 1, \dots, n$). then summing over all i , we **cost** $\sum_{i=1}^n i = \frac{n(n+1)}{2} = O(n^2)$

5 Q5

As we know that we can express $p(x) = \sum_{i=0}^n y_i L_i(x)$ with $L_i(x) = \prod_{j \neq i} \frac{x-x_j}{x_i-x_j}$ to express them into monomial basis then we get : $p(x) = c_0 + c_1x + c_2x^2 + \dots + c_nx^n$. we need to find all coefficient which need to convert the lagrange into monomial form.

We can represent the interpolating polynomial $p(x)$ as: $p(x_k) = y_k$, for all $k = 0, 1, \dots, n$

For $n+1$ interpolation points (x_k, y_k) , substitute the monomial basis representation: $p(x_k) = c_0 + c_1x_k + c_2x_k^2 + \dots + c_nx_k^n$
this lead to a $n+1$ equation:

$$\begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_n \end{bmatrix}$$

this matrix is similar to vandermonde system used for computing the monomial basis coefficients. Thus, the Lagrange interpolant coefficients can also be computed by solving this Vandermonde system.

next we calculate the complexity :

1. we construct $(n+1)^2$ entries Vandermonde matrix requiring $O(n^2)$ flops since each entry involves evaluating powers of x_k
 2. then we solve it require $\frac{2}{3}(n+1)^3$ flops
- we cost the $O(n^3)$ in total.